Tue, Jun 2 at 4:21 AM

how to prove that the sign of \sin^{2}\left(\pi\ x\right)\ \left(\frac{1}{x^{2}}+\frac{1}{\left(1-x\right)^{2}}\right)-8 is positive for x in (0, 1)?

Thought for 55s

The statement as written is **not strictly true** on all of $(0,1)$, because at $x=\frac{1}{2}$,

$$\sin^2(\pi/2)\left(\frac{1}{(1/2)^2}+\frac{1}{(1/2)^2}\right)-8=1(4+4)-8=0.$$

So the correct statement is:

$$\sin^2(\pi x)\left(\frac{1}{x^2}+\frac{1}{(1-x)^2}\right)-8\geq 0$$

for $x\in(0,1)$, with equality only at $x=\frac{1}{2}$. Hence it is positive for $x\neq\frac{1}{2}$.

Set

$$u=2x-1,\qquad u\in(-1,1).$$

Then

$$x=\frac{1+u}{2},\qquad 1-x=\frac{1-u}{2},$$

and

$$\sin(\pi x) = \sin\left(\frac{\pi}{2} + \frac{\pi u}{2}\right) = \cos\left(\frac{\pi u}{2}\right).$$

Also,

$$\frac{1}{x^2} + \frac{1}{(1-x)^2} = \frac{4}{(1+u)^2} + \frac{4}{(1-u)^2} = \frac{8(1+u^2)}{(1-u^2)^2}.$$

Therefore the expression becomes

$$8\left[\frac{(1+u^2)\cos^2\left(\frac{\pi u}{2}\right)}{(1-u^2)^2} - 1\right].$$

So it is enough to prove

$$\frac{(1+u^2)\cos^2\left(\frac{\pi u}{2}\right)}{(1-u^2)^2} \geq 1.$$

Equivalently,

$$\cos\left(\frac{\pi u}{2}\right) \geq \frac{1-u^2}{\sqrt{1+u^2}}, \qquad |u| < 1.$$

Because both sides are even, assume $u \in [0, 1)$. Define

$$F(u) = \log\left(\frac{(1+u^2)\cos^2\left(\frac{\pi u}{2}\right)}{(1-u^2)^2}\right).$$

Then $F(0) = 0$. We show $F'(u) \geq 0$. Differentiating gives

$$F'(u) = -\pi \tan\left(\frac{\pi u}{2}\right) + \frac{2u}{1+u^2} + \frac{4u}{1-u^2}.$$

Use the standard partial fraction expansion

$$\pi \tan\left(\frac{\pi u}{2}\right) = \frac{4u}{1-u^2} + 4u \sum_{n=2}^{\infty} \frac{1}{(2n-1)^2 - u^2}.$$

Hence

$$F'(u) = \frac{2u}{1+u^2} - 4u \sum_{n=2}^{\infty} \frac{1}{(2n-1)^2 - u^2}.$$

For $n \geq 2$ and $0 \leq u < 1$,

$$\frac{1}{(2n-1)^2 - u^2} \leq \frac{1}{2(1+u^2)n(n-1)}.$$

Indeed, this is equivalent to

$$2(1+u^2)n(n-1) \leq (2n-1)^2 - u^2,$$

and the difference is

$$(2n^2 - 2n + 1)(1 - u^2) \geq 0.$$

Therefore

$$\sum_{n=2}^{\infty} \frac{1}{(2n-1)^2 - u^2} \leq \frac{1}{2(1+u^2)} \sum_{n=2}^{\infty} \frac{1}{n(n-1)}.$$

But

$$\sum_{n=2}^{\infty} \frac{1}{n(n-1)} = 1,$$

so

$$\sum_{n=2}^{\infty} \frac{1}{(2n-1)^2 - u^2} \leq \frac{1}{2(1+u^2)}.$$

Thus

$$F'(u) \geq \frac{2u}{1+u^2} - 4u \cdot \frac{1}{2(1+u^2)} = 0.$$

So $F(u) \geq F(0) = 0$. Hence

$$\frac{(1+u^2)\cos^2\left(\frac{\pi u}{2}\right)}{(1-u^2)^2} \geq 1.$$

Therefore

$$\sin^2(\pi x)\left(\frac{1}{x^2} + \frac{1}{(1-x)^2}\right) \geq 8.$$

So

$$\boxed{\sin^2(\pi x)\left(\frac{1}{x^2}+\frac{1}{(1-x)^2}\right)-8\geq 0}$$

for $x \in (0,1)$, with equality only at

$$\boxed{x=\frac{1}{2}.}$$

So the expression is **positive on** $(0,1)\setminus\{\frac{1}{2}\}$, not on the whole interval $(0,1)$.

If you want, I can:

- ↳ Explain why the function is zero at x=1/2 and positive elsewhere
- ↳ Provide an alternative proof using a different variable substitution
- ↳ Summarize the inequality proof focusing on key steps and logic